

§8. TÍNH CHẤT CỦA PHÉP KHAI PHƯƠNG

1) Căn thức bậc hai của một bình phương

$$\sqrt{A^2} = |A| = \begin{cases} A & \text{khi } A \geq 0 \\ -A & \text{khi } A < 0 \end{cases}$$

Ví dụ: Tính: $\sqrt{(2 + \sqrt{3})^2} = |2 + \sqrt{3}| = 2 + \sqrt{3}$ (vì $2 + \sqrt{3} > 0$)

$$\sqrt{(1 - \sqrt{2})^2} = |1 - \sqrt{2}| = -(1 - \sqrt{2}) = -1 + \sqrt{2} \quad (\text{vì } 1 - \sqrt{2} < 0)$$

2) Căn thức bậc hai của một tích

Với $A \geq 0$ và $B \geq 0$, ta có: $\sqrt{A \cdot B} = \sqrt{A} \cdot \sqrt{B}$

Ví dụ: Tính: $\sqrt{36 \cdot 9} = \sqrt{36} \cdot \sqrt{9} = 6 \cdot 3 = 18$.

$$\sqrt{5} \cdot \sqrt{20} = \sqrt{5 \cdot 20} = \sqrt{100} = 10$$

Với số thực a bất kì và $b \geq 0$, ta có:

$$\sqrt{a^2 b} = |a| \sqrt{b} = \begin{cases} a\sqrt{b} & \text{khi } a \geq 0 \\ -a\sqrt{b} & \text{khi } a < 0 \end{cases}$$

Ví dụ: Tính: $\sqrt{12} = \sqrt{4 \cdot 3} = \sqrt{2^2 \cdot 3} = 2\sqrt{3}$ (đưa thừa số ra ngoài dấu căn hay rút gọn căn)

$$4\sqrt{3} = \sqrt{4^2 \cdot 3} = \sqrt{48} \quad (\text{đưa thừa số vào trong dấu căn})$$

3) Căn thức bậc hai của một thương

$$\text{Với } A \geq 0 \text{ và } B > 0, \text{ ta có: } \sqrt{\frac{A}{B}} = \frac{\sqrt{A}}{\sqrt{B}}$$

Ví dụ: Tính: $\sqrt{\frac{81}{16}} = \frac{\sqrt{81}}{\sqrt{16}} = \frac{9}{4}$; $\sqrt{32} : \sqrt{2} = \sqrt{32 : 2} = \sqrt{16} = 4$.

Bài 1. Tính:

- | | | | |
|---|----------------|---|----------------|
| 1) $\sqrt{8}$ | 2) $\sqrt{27}$ | 3) $\sqrt{50}$ | 4) $\sqrt{32}$ |
| 5) $3\sqrt{2} - \sqrt{8} + 2\sqrt{32}$ | | 6) $2\sqrt{5} - \sqrt{125} + \sqrt{80}$ | |
| 7) $2\sqrt{3} - \sqrt{75} + 3\sqrt{12}$ | | 8) $2\sqrt{32} + 3\sqrt{72} - 7\sqrt{50} + \sqrt{2}$ | |
| 9) $6\sqrt{12} - \sqrt{20} - 4\sqrt{27} + \sqrt{125}$ | | 10) $\sqrt{27} - 2\sqrt{3} + 2\sqrt{48} - 3\sqrt{75}$ | |
| 11) $\sqrt{243} - \frac{1}{2}\sqrt{12} - 2\sqrt{75} + 2\sqrt{27}$ | | 12) $\frac{2}{5}\sqrt{75} - 0,5\sqrt{48} + \sqrt{300} - \frac{2}{3}\sqrt{12}$ | |
| 13) $2\sqrt{24} - \frac{3}{5}\sqrt{150} - \sqrt{54} + 4\sqrt{96}$ | | 14) $\frac{1}{2}\sqrt{300} - 5\sqrt{12} + 9\sqrt{27}$ | |

Bài 2. Tính:

- | | |
|--|---|
| 1) $2\sqrt{3}(\sqrt{3} - 2) + 4\sqrt{3}$ | 2) $2\sqrt{5}(\sqrt{5} - \sqrt{10}) + 10\sqrt{2}$ |
| 3) $(5 + \sqrt{5})(\sqrt{5} - 2)$ | 4) $(13 - 4\sqrt{3})(7 + 4\sqrt{3})$ |

5) $(3\sqrt{3} - 4)(2\sqrt{3} + 1)$

7) $(2 + 3\sqrt{5})^2 - 12\sqrt{5}$

9) $(\sqrt{5} - \sqrt{2})^2 + 2\sqrt{10}$

11) $\sqrt{18} \cdot \sqrt{2} - \frac{\sqrt{72}}{\sqrt{2}} + \sqrt{444} : \sqrt{111}$

6) $(2\sqrt{3} + 3)(\sqrt{3} - 1)$

8) $(3 + \sqrt{5})^2 - 8\sqrt{5} - 14$

10) $(\sqrt{2} - \sqrt{3})^2 + 2\sqrt{6}$

12) $(5\sqrt{6} - 4\sqrt{10} + 7\sqrt{30}) : \sqrt{2}$

Bài 3. Rút gọn các biểu thức sau:

1) $\sqrt{(\sqrt{3} - \sqrt{5})^2}$

3) $\sqrt{(4 + \sqrt{15})^2}$

5) $\sqrt{(2\sqrt{3} - 4)^2}$

7) $\sqrt{(\sqrt{2} - \sqrt{3})^2} - \sqrt{(1 - \sqrt{3})^2}$

9) $\sqrt{(2\sqrt{2} - 3)^2} + \sqrt{(2\sqrt{2} - 1)^2}$

11) $\sqrt{(\sqrt{5} - 3)^2} - \sqrt{(2 + \sqrt{5})^2}$

13) $\sqrt{(4 - \sqrt{17})^2} - \sqrt{(\sqrt{17} - 3)^2}$

15) $\sqrt{(3 + \sqrt{11})^2} + \sqrt{(-\sqrt{11} + 3)^2}$

2) $\sqrt{(3 - \sqrt{11})^2}$

4) $\sqrt{(\sqrt{5} - \sqrt{3})^2}$

6) $\sqrt{(3 + \sqrt{8})^2}$

8) $\sqrt{(\sqrt{5} - 2)^2} - \sqrt{(\sqrt{5} - 4)^2}$

10) $\sqrt{(\sqrt{5} - 3)^2} + \sqrt{(2 + \sqrt{5})^2}$

12) $\sqrt{(5 - \sqrt{6})^2} - \sqrt{(2 - \sqrt{6})^2}$

14) $\sqrt{(3 - \sqrt{10})^2} - \sqrt{(2\sqrt{10} - 7)^2}$

16) $\sqrt{(-\sqrt{5} + 3)^2} + \sqrt{(\sqrt{5} + 1)^2}$

Bài 4. Tính:

1) $\sqrt{8 - 2\sqrt{15}}$

3) $\sqrt{9 + 4\sqrt{5}}$

5) $\sqrt{6 + 4\sqrt{2}} - \sqrt{6 - 4\sqrt{2}}$

7) $\sqrt{11 + 2\sqrt{18}} + \sqrt{7 - 2\sqrt{10}}$

9) $\sqrt{3 + 2\sqrt{2}} - \sqrt{5 + 2\sqrt{6}}$

11) $\sqrt{23 - 8\sqrt{7}} + \sqrt{10 - 2\sqrt{21}}$

13) $\sqrt{8 - 2\sqrt{15}} - \sqrt{23 - 4\sqrt{15}}$

15) $\sqrt{10 - \sqrt{84}} + \sqrt{34 - 2\sqrt{189}}$

2) $\sqrt{3 + 2\sqrt{2}}$

4) $\sqrt{15 - 6\sqrt{6}}$

6) $\sqrt{18 - 2\sqrt{21}} - \sqrt{9 - 2\sqrt{14}}$

8) $\sqrt{6 - 4\sqrt{2}} + \sqrt{3 + 2\sqrt{2}}$

10) $\sqrt{11 - 6\sqrt{2}} - \sqrt{3 - 2\sqrt{2}}$

12) $\sqrt{7 - 4\sqrt{3}} - \sqrt{12 - 6\sqrt{3}}$

14) $\sqrt{13 - 4\sqrt{10}} + \sqrt{7 - 2\sqrt{10}}$

16) $\sqrt{13 - 4\sqrt{10}} - \sqrt{53 + 12\sqrt{10}}$

Bài 5. Tính:

1) $\sqrt{7 - 2\sqrt{10}} - \sqrt{(\sqrt{2} - \sqrt{5})^2}$

3) $\sqrt{(4 - 3\sqrt{3})^2} + \sqrt{28 - 16\sqrt{3}}$

5) $\sqrt{19 - 8\sqrt{3}} - \sqrt{(1 - 2\sqrt{3})^2}$

7) $\sqrt{27 - 10\sqrt{2}} - \sqrt{(3\sqrt{2} - 5)^2}$

2) $\sqrt{11 - 6\sqrt{2}} + \sqrt{(1 - \sqrt{2})^2}$

4) $\sqrt{(2 - \sqrt{5})^2} + \sqrt{14 - 6\sqrt{5}}$

6) $\sqrt{(3 + \sqrt{2})^2} + \sqrt{6 - 4\sqrt{2}}$

8) $\sqrt{4 - 2\sqrt{3}} + \sqrt{(2 - \sqrt{3})^2}$

$$9) (2 - \sqrt{3})\sqrt{7 + 4\sqrt{3}}$$

$$10) (3 - \sqrt{2})\sqrt{11 + 6\sqrt{2}}$$

$$11) \sqrt{10 + 2\sqrt{21}}(\sqrt{3} - \sqrt{7})$$

$$12) (\sqrt{3} - \sqrt{5})\sqrt{8 + \sqrt{60}}$$

$$13) \sqrt{15 + 6\sqrt{6}}(3 - \sqrt{6})$$

$$14) \sqrt{19 - 6\sqrt{2}} \cdot (1 + 3\sqrt{2})$$

$$15) \sqrt{3 + \sqrt{29 + 12\sqrt{5}}}$$

$$16) \sqrt{17 - 6\sqrt{2 + \sqrt{9 + 4\sqrt{2}}}}$$

Đs: 1) 0 ; 2) 2 ; 3) $\sqrt{3}$; 4) 1 ; 5) $5 - 5\sqrt{3}$; 6) 5 ; 7) $2\sqrt{2}$; 8) 1 ; 9) 1 ; 10) 7 ; 11) -4 ; 12) -2 ; 13) 3 ; 14) 17 ; 15) $\sqrt{5} + 1$; 16) $3 - \sqrt{2}$.

Bài 6. Tính (Khai phương căn thiếu):

$$1) \sqrt{2} \cdot (\sqrt{7 - 3\sqrt{5}} + \sqrt{7 + 3\sqrt{5}})$$

$$2) \sqrt{2} \cdot (\sqrt{2 - \sqrt{3}} - \sqrt{2 + \sqrt{3}} + \sqrt{2})$$

$$3) \sqrt{2} \cdot \sqrt{7 + 3\sqrt{5}} - (\sqrt{5} + 1)$$

$$4) \sqrt{2} \cdot (\sqrt{6 - \sqrt{35}} - \sqrt{6 + \sqrt{35}})$$

$$5) (\sqrt{10} - \sqrt{2})\sqrt{3 + \sqrt{5}}$$

$$6) (3\sqrt{2} + \sqrt{10})\sqrt{7 - \sqrt{45}}$$

$$7) \sqrt{4 + \sqrt{15}}(\sqrt{6} - \sqrt{10})$$

$$8) \sqrt{7 + 3\sqrt{5}}(\sqrt{10} - 3\sqrt{2})$$

$$9) \sqrt{5 - \sqrt{21}}(\sqrt{14} + \sqrt{6})$$

$$10) (\sqrt{14} - \sqrt{6})\sqrt{5 + \sqrt{21}}$$

$$11) (\sqrt{6} + \sqrt{2})\sqrt{2 - \sqrt{3}}$$

$$12) \sqrt{3 + \sqrt{5}}(\sqrt{2} - \sqrt{10})$$

$$13) \sqrt{(\sqrt{10} + \sqrt{6})\sqrt{4\sqrt{15} + 2}}$$

$$14) \sqrt{7 - \sqrt{33}}(\sqrt{22} + \sqrt{6})$$

$$15) (\sqrt{3 - \sqrt{5}} + \sqrt{2 + \sqrt{3}})(\sqrt{10} - \sqrt{6})$$

$$16) (\sqrt{11 - 6\sqrt{2}} - \sqrt{15 - 6\sqrt{6}})\sqrt{2 + \sqrt{3}}$$

$$17) \left(\sqrt{2 + \sqrt{3}} + \sqrt{3 - \sqrt{5}} - \frac{\sqrt{5}}{\sqrt{2}} \right)^2$$

$$18) \left(\sqrt{2 - \sqrt{3}} + \sqrt{3 + \sqrt{5}} - \sqrt{\frac{3}{2}} \right)^2$$

$$19) \frac{\sqrt{29 + 12\sqrt{5}} - \sqrt{29 - 12\sqrt{5}}}{(3\sqrt{2} + \sqrt{10})\sqrt{7 - 3\sqrt{5}}}$$

$$20) \left(\sqrt{4 + \sqrt{7}} + \sqrt{2 - \sqrt{3}} - \sqrt{\frac{7}{2}} \right)^2$$

Đs: 1) 6 ; 2) 0 ; 3) 2 ; 4) $-2\sqrt{5}$; 5) 4 ; 6) 4 ; 7) -2 ; 8) -4 ; 9) 4 ; 10) 4 ; 11) 2 ; 12) -4 ; 13) 2 ; 14) 8 ; 15) 2 ; 16) 2 ; 17) $\frac{3}{2}$; 18) $\frac{5}{2}$; 19) $\frac{3}{2}$; 20) $\frac{3}{2}$.

